

CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

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ABSTRACT

Adversarial training is among the most reliable defenses against adversarial examples, but it degrades standard accuracy, and this trade-off remains the central obstacle to its practical use. The accuracy that is lost is not lost in general: a network of the same architecture, trained naturally on the same data, still holds it. We therefore ask whether that accuracy can be put back by self-distillation, using the model’s own naturally trained counterpart—available at no adversarial cost—for the single purpose of restoring what adversarial training gives up, while robustness continues to be produced entirely by the inner maximization. Such a teacher is fragile—under $\epsilon = 8/255$ its own feature rotates by 63.8° —and consuming it as adversarial distillation does makes matters worse: every published objective given this teacher lands below simply not distilling. We show that the fragility is not transferred when the teacher is used as a *feature anchor* instead. Matching the teacher’s clean feature at the adversarial input means the teacher is never evaluated under attack, so it transfers neither its instability nor any robustness, and we prove that this single term is equivalent within a constant factor to fidelity plus local stability, with no coefficient between them. Combined with a per-sample attack radius set by the input sensitivity of the loss at fixed total budget, the method has no loss weight, no temperature and no trained classifier, and transfers across datasets unchanged. On CIFAR-100 with ResNet-18 it improves standard accuracy by 5.3 points at no cost in AutoAttack, and it sets a new accuracy-robustness frontier on CIFAR-10, CIFAR-100 and Tiny-ImageNet. Because the teacher is naturally trained, its training length becomes a trade-off control that costs nothing to exercise.

1 INTRODUCTION

Deep neural networks are vulnerable to adversarial examples, imperceptible perturbations of the input that change the prediction (Goodfellow et al., 2014; Madry et al., 2017; Carlini & Wagner, 2017), which raises concerns wherever such models are deployed in safety-critical systems (Ma et al., 2021; Grigorescu et al., 2020). Among the defenses proposed in response, adversarial training (Madry et al., 2017) remains the most reliable (Athalye et al., 2018; Croce & Hein, 2020): the model is optimized on perturbations produced by an inner maximization, and the robustness it attains has survived a decade of stronger attacks. Its cost, however, is standard accuracy. A network trained adversarially classifies unperturbed inputs considerably worse than the same network trained naturally, and the gap grows with the perturbation budget. This accuracy-robustness trade-off is partly intrinsic to the objective (Tsipras et al., 2018; Zhang et al., 2019) and it is the principal obstacle to deploying adversarially trained models in practice.

Two lines of work reduce it. The first modifies the target inside adversarial training. Hard labels are a poor supervisory signal for perturbed inputs, since they assert that every point of an ϵ -ball belongs to one class with full confidence, and softening them alleviates both the trade-off and robust overfitting (Rice et al., 2020). Label smoothing (Pang et al., 2020), consistency regularization (Tack et al., 2022), and annealing self-distillation rectification (Wu et al., 2024) all replace the one-hot

target with a smoother one, the last using the model’s own weight average as the source. These methods require no additional network. The second line is adversarial distillation, which supplies the target from an adversarially trained teacher: ARD (Goldblum et al., 2020), IAD (Zhu et al., 2021), RSLAD (Zi et al., 2021), AdaAD (Huang et al., 2023) and IGDM (Lee et al., 2025) differ in which teacher quantity is matched and where the teacher is evaluated, and together they define the strongest reported accuracy–robustness frontiers for small architectures.

The two lines leave a gap between them. Adversarial distillation presupposes that an already-robust network exists, and in practice that network is a WideResNet adversarially trained with large volumes of generated data (Wang et al., 2023)—costlier to obtain than the student it teaches. The robustness such methods report therefore originates outside the student, and the contribution being measured is the quality of the transfer. The teacher that is genuinely free, a naturally trained copy of the student’s own architecture, is not used as a source of guidance for the adversarial term, and the reason is well founded: a naturally trained network attains its accuracy through features an adversary can manipulate (Ilyas et al., 2019), so transferring what it computes is expected to transfer that fragility as well. ARREST (Suzuki et al., 2023) comes closest to using one—it initializes the student from a standardly pretrained network and penalizes the angular distance between the student’s representation of an adversarial example and the pretrained network’s representation of the clean example—but there the penalty is added to the label cross-entropy as one weighted term among several, so the backbone is still driven by hard labels and the balance between the two must be tuned. DP-FAT (Cho & Kim, 2026) distills a naturally trained teacher directly, in the fast single-step regime, and reports that the teacher’s logit *magnitudes* are the unreliable part, rescaling them per sample so that only the class-discriminative direction is transferred. That is the closest existing use of a natural teacher, and the difference from ours is what the teacher is asked for: DP-FAT repairs a logit target so that it can be trusted, where we remove the logit target and place the anchor on the feature, which is where the magnitude question does not arise. B-MTARD (Zhao et al., 2024) goes further still and uses a naturally trained network explicitly to recover accuracy, pairing a clean teacher with a robust one so that each handles the examples it is good at. That is the closest prior statement of the idea we pursue, and it still requires the robust teacher—a WideResNet-70-16 on CIFAR-100, far larger than the student—because the clean teacher supplies logits and cannot supply robustness.

We do not claim that a naturally trained teacher is preferable to a robust one, and we do not compete with methods that have one. A robust teacher is better guidance for robustness, and obtaining it is precisely the expensive part. Our question is narrower and is about the other axis: adversarial training discards standard accuracy that a naturally trained copy of the same network still holds, and we ask whether self-distillation can put it back. The answer turns on *what the teacher is asked to supply*. We ask it for standard accuracy alone and leave robustness entirely to the student’s inner maximization, and under that division the expected fragility does not materialize—for a structural reason rather than an empirical one. Our objective matches the teacher’s feature at the *clean* input against the student’s feature at the *adversarial* input, and it is the entire objective for the backbone: the label term is removed rather than supplemented. The teacher is therefore never evaluated inside the perturbation ball, so nothing it does there can enter the optimization, and by the same token it cannot contribute robustness. All robustness is produced by the student’s inner maximization, exactly as in standard adversarial training. We further show that this single term is not merely a fidelity constraint: writing F for the student’s clean deviation from the teacher and O for the diameter of the student’s own output over the ϵ -ball, the anchor loss L satisfies $L \leq F + O \leq 3L$, so minimizing it controls teacher fidelity and local stability together, within a constant factor, with no weight to balance them. The bound uses only the triangle inequality and therefore holds for the networks we actually train.

A second finding concerns what the teacher supplies. Across a ladder of naturally trained teachers differing only in training length, teacher accuracy and student accuracy are almost perfectly *anti*-correlated ($r = -0.999$): the most accurate teacher produces the worst student. What the student inherits is the teacher’s class geometry rather than its predictions, and the two are not monotone in one another. This makes the teacher’s training length a control on the accuracy–robustness frontier that costs nothing to exercise and that a robust teacher does not offer, since a robust teacher’s own training is the expensive part. The non-monotonicity itself is not peculiar to natural teachers: in the robust-teacher setting a more robust teacher is likewise reported to fail to improve, or to harm, the student, with the teacher’s predictive entropy on a consistent subset of the data indicating which way

it will go (Anonymous, 2026). Our ladder is the natural-teacher analogue, and what varies along it is class separation rather than accuracy.

Building on these observations we propose **Clean Feature Anchoring** (CFA). CFA trains a naturally supervised copy of the student architecture on the same data, initializes the student from it, and then trains the backbone with the clean-feature anchor alone; the classifier is inherited and never trained. To this we add **sensitivity-matched** ϵ , which redistributes the per-sample attack radius in inverse proportion to the input sensitivity of the loss while holding the total budget fixed, so that each example is perturbed to a comparable displacement of the objective rather than a comparable displacement of the input. The resulting method has no temperature, no loss weight and no auxiliary head, and we apply an identical configuration to every dataset, changing only the dataset name and the teacher checkpoint.

On CIFAR-100 with ResNet-18, CFA reaches 62.65% standard accuracy at 28.77% AutoAttack accuracy. Against the strongest method at comparable robustness this is +5.29 points of standard accuracy, and against the strongest method at comparable standard accuracy it is +4.81 points of AutoAttack accuracy; no method in this setting exceeds ours on both axes. We further run the published distillation objectives on our own natural teacher, which is the teacher they would have in this setting: ARD, RSLAD and AdaAD reach 20.24%, 21.30% and 23.19% AutoAttack accuracy, all below plain adversarial training initialized at the same teacher (26.46%). The accuracy a natural teacher holds is therefore not recovered by treating it as a logit target. Our contributions are as follows:

- We show that a naturally trained network can guide adversarial training when it is asked only for standard accuracy. Evaluated at the clean input and used as the entire backbone objective, a feature anchor controls fidelity to the teacher and local stability of the student jointly, within a constant factor, as a single unweighted term—so the teacher’s own fragility has no path into the optimization, and no coefficient trades the two properties off.
- We show that what a naturally trained teacher transfers is class geometry rather than accuracy, with teacher and student accuracy anti-correlated at $r = -0.999$ across teachers, which turns the teacher’s training length into a trade-off control available only in this setting.
- We propose Clean Feature Anchoring together with sensitivity-matched ϵ , a method with no loss weight, no temperature and no per-dataset tuning, in which robustness remains entirely attributable to the student’s own inner maximization.
- We demonstrate a new accuracy–robustness frontier on CIFAR-10, CIFAR-100 and Tiny-ImageNet, and show that the gain is not an artifact of weight averaging or adversarial weight perturbation: with weight averaging alone and half the training schedule, CFA already exceeds the best published result that uses both. Running the published distillation objectives on the same natural teacher places all of them below plain adversarial training from that teacher, which locates the difficulty in the logit target rather than in the teacher.

2 ANALYSIS

2.1 PRELIMINARIES

Let $f = h \circ \Phi$ be a classifier with feature map Φ and linear head h . Adversarial training (Madry et al., 2017) solves

$$\min_{\theta} \mathbb{E}_{(x,y)} \max_{x' \in B(x,\epsilon)} \ell(f_{\theta}(x'), y), \quad B(x, \epsilon) = \{x' : \|x' - x\|_{\infty} \leq \epsilon\}. \quad (1)$$

The supervisory signal is the label, and it is asserted uniformly over $B(x, \epsilon)$: every point of the ball must reach full confidence in one class. That is where the accuracy cost comes from. We write Φ_t for the frozen feature map of a network of the same architecture trained naturally on the same data—the teacher throughout—and ask what it can supply.

Table 1: Published adversarial-distillation objectives given a *naturally* trained teacher. CIFAR-100, ResNet-18, 100 epochs, each method at its published recipe. Every one lands below simply not distilling.

	clean	AA	NRR
ARD (Goldblum et al., 2020)	57.61	20.24	29.96
RSLAD (Zi et al., 2021)	59.68	21.30	31.40
AdaAD (Huang et al., 2023)	59.79	23.19	33.42
PGD-AT, initialized at the teacher	57.73	26.46	36.29

2.2 THE TEACHER IS FRAGILE, AND ITS LOGITS ARE WORSE THAN NO TEACHER AT ALL

The objection to a naturally trained teacher is not hypothetical. Under $\epsilon = 8/255$ its own feature rotates by 63.8° and its norm inflates by $\times 2.45$; its accuracy rests on directions an adversary controls.

That fragility appears immediately if the teacher is consumed the way adversarial distillation consumes one. Table 8 gives the published objectives with our natural teacher in place of the robust teacher they assume, each at its own published recipe, against plain adversarial training initialized from the same teacher.

None of it is a temperature that was left untuned: swept in a fixed base regime, the logit target peaks at 24.48 AutoAttack ($\tau = 4$) and the curve is unimodal because τ is squeezed from both sides—at $\tau = 1$ the teacher’s maximum probability is 0.820 against the student’s 0.016, and at $\tau = 16$ the target is within 1% of uniform.

Two defects, and they separate. A method can go wrong in *where* it reads the teacher and in *what* it reads from it, and Table 8 confounds them. Holding the read point fixed at x and changing only what is read—the teacher’s logits against its features, same regime, same attack, same teacher—gives $24.48 \rightarrow \mathbf{25.88}$ AutoAttack. That isolates the second defect.

The first is larger, and is visible where a method reads the teacher *off* the clean point. IGDM (Lee et al., 2025) matches a finite difference $f_T(x + \delta) - f_T(x - \delta)$ as a stand-in for the teacher’s input gradient, which is a gradient only while the teacher is locally linear; its own diagnostic reports a Taylor remainder proportion of 0.012 for adversarially trained teachers. Applying that diagnostic to our checkpoints gives 0.0111 for an adversarially trained ResNet-18—reproducing their value, so the measurement is calibrated—and **0.4763** for the natural teacher, forty times further from linear. The target it constructs is correspondingly degenerate: $\|f_T(x + \delta) - f_T(x - \delta)\| = 14.14$ against $\|f_T(x)\| = 18.36$, with the softmax of that difference placing 0.917 on a single class.

The teacher therefore holds accuracy that adversarial training gives up, and the logit route does not recover it. What follows keeps the read point at x , where the first defect cannot arise, and takes the features, which the isolation above shows to be the better read.

2.3 WHAT A CLEAN-FEATURE ANCHOR CONTROLS

Consider matching the teacher’s feature at the *clean* input against the student’s feature at the *adversarial* input, that is, minimizing

$$\mathcal{L}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|^2. \quad (2)$$

Read literally, Equation (2) asks the student to reproduce a *fixed* vector from every point of the ball. That is a fidelity constraint, and a fidelity constraint alone would have no reason to produce robustness. It turns out not to be one, and the reason is visible once the quantity is compared with the two properties one would otherwise have to impose separately.

Fix x , write $B = B(x, \epsilon)$, and define

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|, \quad (3)$$

$$F = \|\Phi_s(x) - \Phi_t(x)\|, \quad (4)$$

$$O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|. \quad (5)$$

Here L is the square root of Equation (2), the quantity the method minimizes; F is *fidelity*, how far the student is from the teacher at the clean point; and O is *oscillation*, the diameter of the student's own feature map over the ball. F is what adversarial training gives up relative to natural training, and O is what its label term buys. They are ordinarily two objectives.

Proposition 1. *For every x and every $\epsilon \geq 0$,*

$$L \leq F + O \leq 3L. \quad (6)$$

Proof. For the right-hand inequality, $F \leq L$ because $x \in B$, so the maximum in Equation (3) is at least the value at x . For O , insert $\Phi_t(x)$ between the two student features: for any $x', x'' \in B$,

$$\|\Phi_s(x') - \Phi_s(x'')\| \leq \|\Phi_s(x') - \Phi_t(x)\| + \|\Phi_t(x) - \Phi_s(x'')\| \leq 2L,$$

and taking the maximum over x', x'' gives $O \leq 2L$. Adding the two bounds gives $F + O \leq 3L$. For the left-hand inequality, insert $\Phi_s(x)$ instead: for any $x' \in B$,

$$\|\Phi_s(x') - \Phi_t(x)\| \leq \|\Phi_s(x') - \Phi_s(x)\| + \|\Phi_s(x) - \Phi_t(x)\| \leq O + F,$$

and taking the maximum over x' gives $L \leq F + O$. $\square$

Only the triangle inequality is used, so Proposition 1 holds for the networks we actually train: no smoothness, no linearity, no assumption on the architecture or on the attack that realizes the maximum. Two consequences follow immediately.

If the anchor is driven to $L \leq \delta$ at x , then $F \leq \delta$ and $O \leq 2\delta$ at x .

That is the operative statement: driving one scalar down bounds *both* properties, at rates that differ by a factor of two and by nothing else. Conversely, by the left inequality, L cannot be small unless both are, so the objective admits no solution that satisfies one at the expense of the other. A weighted sum $F + \lambda O$ has no such property for any fixed λ : its level sets contain points where either term is arbitrarily large. This is why the anchor is stated as the whole objective rather than as a regularizer, and why there is no coefficient to select.

Where the constant 3 comes from, and how tight it is. The factor is $1 + 2$: one from $F \leq L$ and two from $O \leq 2L$, and the second is attained when the student's outputs at the two extreme points of the ball sit on opposite sides of $\Phi_t(x)$ at distance L each. Our trained students are far from that configuration, which is the content of the measurement below.

The teacher appears at x and nowhere else. Equation (2) evaluates Φ_t at the clean point only, in the loss and in the attack that realizes the maximum. Whatever the teacher does inside B is therefore absent from the objective, and the instability measured in Section 2.2—63.8° of rotation, $\times 2.45$ of norm—has no path into the optimization. The same fact settles the symmetric question: since the teacher is never asked to resist a perturbation, it cannot supply robustness either, and all of it is produced by the student's inner maximization. Section 4.4 tests this by moving the read point and changing nothing else.

Verification. Proposition 1 predicts that a student trained on L ends up with a small O , and the prediction is checkable on the trained network. Measured over $\epsilon = 8/255$ balls on CIFAR-100, $L = 7.58$, $F = 6.30$ and $O_{\text{lb}} = 2.16$, against the teacher's own $O = 6.68$: the student is $3.1 \times$ more stable than the network it was distilled from, on the same balls. Two things follow. The teacher's instability was not inherited, which is what the construction predicts. And $F \gg O_{\text{lb}}$, so what remains in the loss at convergence is fidelity rather than instability—the anchor is not being satisfied by making the student flat.

What Proposition 1 does not establish. It bounds $F + O$, and O is not a sufficient statistic for robust accuracy: across the teacher ladder of Section 4.6 it correlates with AutoAttack at $r = +0.83$, the wrong sign. The proposition explains why anchoring to a non-robust teacher is *free*; it is not a robustness bound and we do not present it as one. Section A gives two partial results on the complementary question, each with the reason it falls short.

3 METHOD

3.1 CLEAN FEATURE ANCHORING

Let Φ_t be the frozen feature map of a network of the same architecture trained naturally on the same data, and let h_t be its classifier. The student’s feature map is initialized at $\Phi_s \leftarrow \Phi_t$ and trained with

$$\mathcal{L} = \mathbb{E}_x \|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|^2, \quad x_{\text{adv}} = \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|, \quad (7)$$

where the inner maximization is realized by K steps of projected gradient ascent on the same quantity the outer problem minimizes,

$$x^{(k+1)} = \Pi_{B(x, \epsilon)} \left(x^{(k)} + a \cdot \text{sign} \nabla_x \|\Phi_s(x^{(k)}) - \Phi_t(x)\|^2 \right), \quad x^{(0)} = x + \eta, \quad (8)$$

with η a small random initialization and Π the projection onto the ball. Inner and outer problems therefore agree, as they do in standard adversarial training and unlike methods whose attack is generated against a different objective from the one being trained.

Equation (7) is the *entire* objective for the backbone. No label term, no logit term, and no coefficient joining them. Three properties follow from that form, and each is a statement about the construction rather than an empirical observation.

(i) The teacher never enters the perturbation ball. In both Equation (7) and Equation (8) the teacher is evaluated at x alone. Formally, the objective is a functional of Φ_t only through the single vector $\Phi_t(x)$, so two teachers agreeing at x and differing arbitrarily on $B(x, \epsilon) \setminus \{x\}$ induce the same optimization problem. The teacher’s 63.8° of rotation under attack is exactly such a difference, and it is therefore invisible to training. The converse holds by the same argument: a quantity the objective cannot see cannot be a source of robustness either.

(ii) The objective is exactly zero at initialization. Since $\Phi_s \leftarrow \Phi_t$,

$$\mathcal{L}|_{\Phi_s = \Phi_t} = \mathbb{E}_x \|\Phi_t(x_{\text{adv}}) - \Phi_t(x)\|^2 \neq 0 \text{ in general,} \quad \text{but} \quad \|\Phi_s(x) - \Phi_t(x)\| = 0, \quad (9)$$

so the loss is nonzero only because of the perturbation: every gradient the backbone receives at the start of training is attributable to the attack, which is the quantity to be trained. A logit target has this property only at temperature 1, where it carries no dark knowledge. It also has a practical consequence used in Section 3.2: at the first step the per-sample sensitivities are identical, so the radius allocation begins uniform and differentiates only as the student moves.

(iii) The classifier is inherited, and this is what a feature target buys. At test time the student is $h_t \circ \Phi_s$: the head is never trained. This is available because the target is written on the feature. A loss written on logits constrains only the composition $W\Phi$, so it determines the pair (Φ, W) up to the gauge

$$(\Phi, W) \mapsto (A\Phi, WA^{-1}), \quad A \in \text{GL}(d), \quad (10)$$

and the same logits are reachable from a continuum of representations, none of which the teacher’s classifier is calibrated for. Fixing the target at the feature removes that freedom and pins Φ_s to Φ_t itself, so h_t remains the right classifier for it. Table 11 shows this is not merely permissible but preferred: every alternative we tried, including the best head temperature, scores below leaving the classifier alone. It is also what removes the last two hyperparameters, since a head distillation term would reintroduce a temperature and a weight.

3.2 SENSITIVITY-MATCHED ϵ

The problem a uniform radius creates. Equation (7) is optimized over a *pixel* ball while what it measures is a *feature* displacement, and the two are not proportional. The same ϵ moves the objective by an amount that varies across the dataset, and by a lot: logged over training, the coefficient of variation of $\|\nabla_x \mathcal{L}\|$ across a minibatch is 0.64. A fixed radius therefore delivers a strongly non-uniform amount of pressure—some examples are attacked hard in the geometry the loss lives in, others barely at all—even though every example is allotted the same pixel budget. Nothing in Equation (7) asks for that.

Equalizing the movement. Inside an ℓ_∞ ball of radius e the first-order change of any differentiable loss is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (11)$$

since ℓ_1 is the dual norm of ℓ_∞ . Writing $g_i = \|\nabla_x \mathcal{L}(x_i)\|$, the movement induced at example i by a radius ϵ_i is $\epsilon_i g_i$ to first order, so equalizing it across a minibatch of N examples means imposing

$$\epsilon_i g_i = c \quad \text{for all } i, \quad \text{subject to} \quad \sum_{i=1}^N \epsilon_i = N\epsilon. \quad (12)$$

The constraint is what makes the rule comparable to a uniform one: it holds the *total* attack budget fixed. Solving Equation (12) gives $c = N\epsilon / \sum_j g_j^{-1}$ and hence

$$\epsilon_i = N\epsilon \cdot \frac{g_i^{-1}}{\sum_j g_j^{-1}}, \quad \text{generalized to} \quad \epsilon_i \propto g_i^{-p}, \quad (13)$$

with $p = 0$ recovering the uniform radius and $p = 1$ the full equalization above. Equivalently Equation (13) is the max–min allocation: moving budget from the example whose objective moves most to the one whose objective moves least always improves the worst case, and the optimum is where no such move remains.

Keeping the radii in range. Left alone, Equation (13) assigns unbounded radii to examples where $g_i \rightarrow 0$, which happens at the start of training by property (ii) above. We therefore constrain $\epsilon_i/\epsilon \in [w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$ and solve the budget constraint inside the box: with $r_i = (\bar{g}/g_i)^p$, the exact solution is the scale t with

$$\sum_{i=1}^N \text{clip}(t r_i, w_{\text{lo}}, w_{\text{hi}}) = N, \quad \epsilon_i = \epsilon \cdot \text{clip}(t r_i, w_{\text{lo}}, w_{\text{hi}}). \quad (14)$$

The left side is continuous and nondecreasing in t from Nw_{lo} to Nw_{hi} , so the root exists and is found by bisection; when no clip is active it reduces to restoring the mean. We use $p = 1$ throughout and never tune it, and Table 11 reports the surrounding values.

Why this is not the existing per-sample radius. Assigning a different ϵ per example is not new: IAAT (Balaji et al., 2019), MMA (Ding et al., 2020) and CAT (Cheng et al., 2022) all do it. They assign it from the *difficulty* of the example or from its margin in **input** space. Equation (14) assigns it from the input-sensitivity of the **training loss**—the geometry the objective is actually written in—and the two are not the same signal. Measured on the same batches, the difficulty score has a coefficient of variation of 0.03 where the sensitivity has 0.64: allocating by difficulty is close to allocating uniformly.

The distinction is testable without confounding it with the budget. Holding the exact multiset of weights Equation (14) produces—same values, same clip, same mean, therefore the same total attack budget—and only reassigning which example receives which, ordered by difficulty as IAAT and CAT would, costs 0.74 AutoAttack and lands below the uniform baseline on both CW and NRR (Table 11). The same weights in a different order move *along* the frontier; only this order moves the frontier itself.

Algorithm 1 Clean Feature Anchoring (CFA)

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1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , box  $[w_{\text{lo}}, w_{\text{hi}}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \|\nabla_x \|\Phi_s(x_i) - \phi_i\|^2\|$  {one backward pass}
13:      $r_i \leftarrow (\bar{g}/g_i)^p$ ; solve  $\sum_i \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N$  for  $t$  {Equation (14)}
14:      $\epsilon_i \leftarrow \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$   $\{\sum_i \epsilon_i = N\epsilon\}$ 
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|^2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for
```

3.3 ALGORITHM

Phase 1 is the only place labels appear, and it is ordinary supervised training. Phase 2 uses none: the teacher vector ϕ_i is the entire supervisory signal, it is read once per example at the clean input, and the classifier h_t is carried through untouched. The method introduces no loss weight, no temperature and no auxiliary head, and the only quantity we vary between datasets is the teacher checkpoint.

Caveats. Equation (11) asks for the ℓ_1 norm and our runs use ℓ_2 ; both were trained and agree within noise, so the derivation is printed as stated. And g_i is a single backward pass at the clean x_i , so it is the sensitivity at the attack’s starting point rather than along its trajectory; Equation (12) is first-order in ϵ throughout.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETTINGS

Datasets. CIFAR-10, CIFAR-100 (Krizhevsky et al., 2009) and Tiny-ImageNet-200 (Le & Yang, 2015). Random crop with 4-pixel padding and random horizontal flip; no other augmentation.

Teacher. For each dataset, a network of the same architecture as the student, trained naturally for 200 epochs on the same data (Table 2). No adversarial training is performed at any point in the teacher’s construction.

Training details. ResNet-18 throughout the main tables, WideResNet-34-10 in Section 4.7. The student is initialized from the teacher and trained for 100 epochs with AdamW at learning rate 0.021 under a one-cycle schedule, batch size 128. The attack is 10-step PGD at step size $2/255$ with a training radius of $8.8/255$; Section 4.5 varies that radius. Weight averaging ($\kappa = 0.999$, from 20% of the run) and an AWP proxy (Wu et al., 2020) ($\gamma = 0.005$, after epoch 10) are used and ablated in Section 4.4. Every dataset uses this identical configuration; only the dataset name and the teacher checkpoint change.

Table 2: Teacher models. AutoAttack accuracy is 0 by construction: these networks never see an adversarial example.

Dataset	Teacher	Epochs	Clean	AA
CIFAR-10	ResNet-18, natural	200	95.32	0.00
CIFAR-100	ResNet-18, natural	200	77.66	0.00
Tiny-ImageNet	ResNet-18, natural	200	66.29	0.00

Table 3: CIFAR-10, ResNet-18, ℓ_∞ , $\epsilon = 8/255$. **Every row is measured in our framework**—same data pipeline, same attack implementation, same evaluation, final-epoch weight-averaged model—so the numbers are comparable to each other rather than across papers. **Teacher** states what each method assumes: – none, *self* its own weight average, *natural* a naturally trained network. Rows we have not run are left empty with the reason; published numbers for the same methods are given separately in Table 15 and are not mixed in here. Avg = $\frac{1}{2}(\text{Clean} + \text{AA})$; NRR is the harmonic mean. Per-attack metrics are in Table 14.

Method	Teacher	Clean	AA	Avg	NRR	
PGD-AT	–	–	–	–	–	<i>vanilla; only the teacher-init variant is run</i>
PGD-AT, initialized at the teacher	natural	82.23	50.72	66.47	62.74	
TRADES	–	–	–	–	–	<i>not implemented in our framework</i>
MART	–	–	–	–	–	<i>not implemented in our framework</i>
ARD	natural	84.58	43.75	64.16	57.67	
RSLAD	natural	85.15	42.91	64.03	57.06	
AdaAD	natural	85.39	46.22	65.81	59.98	
AdaAD + IGDM ($\alpha = 20$)	natural	10.00	10.00	10.00	10.00	<i>collapses</i>
AdaAD + IGDM ($\alpha = 1$)	natural	–	–	–	–	<i>running</i>
ADR	self	–	–	–	–	<i>external repo present, not yet run</i>
CURE	self	–	–	–	–	<i>seven runs, none reproduce; see Section D</i>
RPAT++	–	82.41	50.75	66.58	62.82	
CFA (ours)[†]	natural	85.58	51.79	68.69	64.53	

Comparison models. Two groups: adversarial training methods requiring no external network (PGD-AT, TRADES, MART, Consistency-AT, RPAT, ADR, CURE, Generalist++, ARREST), and adversarial distillation methods, which assume an adversarially trained teacher (ARD, RSLAD, AdaAD, IGDM, B-MTARD). Published numbers are quoted with their source; RPAT and CURE we additionally re-ran, and the four distillation objectives we ported and ran on our own natural teacher (Section 4.3).

Evaluation. Clean accuracy, FGSM, PGD-20/10/50, CW_∞ and AutoAttack (standard version) on the full test set at $\epsilon = 8/255$. NRR is the harmonic mean of clean and AutoAttack accuracy. All of our numbers are the final-epoch weight-averaged model, with no best-checkpoint selection.

4.2 MAIN RESULTS

Baselines are grouped by what they assume is available: **(a)** standard adversarial training; **(b)** adversarial distillation from an *adversarially trained* teacher; **(c)** methods guided by a *naturally* trained network, the family ours belongs to; and **(d)** current accuracy–robustness trade-off methods. Published numbers carry their source; ‘–’ means the paper does not report that cell for this dataset and architecture.

Adversarial distillation is deliberately absent here. ARD, RSLAD, AdaAD and IGDM report their results from a WideResNet-28-10 or WRN-70-16 teacher adversarially trained with generated data—a network more expensive than the student it teaches—so their robustness originates outside the student and is not comparable to a row produced without one. Placing those numbers in the same column as ours would compare methods that assume different inputs. Section 4.3 gives them

Table 4: CIFAR-100, ResNet-18, ℓ_∞ , $\epsilon = 8/255$. **Every row is measured in our framework**—same data pipeline, same attack implementation, same evaluation, final-epoch weight-averaged model—so the numbers are comparable to each other rather than across papers. **Teacher** states what each method assumes: *–* none, *self* its own weight average, *natural* a naturally trained network. Rows we have not run are left empty with the reason; published numbers for the same methods are given separately in Table 15 and are not mixed in here. Avg = $\frac{1}{2}(\text{Clean} + \text{AA})$; NRR is the harmonic mean. Per-attack metrics are in Table 14.

Method	Teacher	Clean	AA	Avg	NRR	
PGD-AT	–	–	–	–	–	<i>vanilla; only the teacher-init variant is run</i>
PGD-AT, initialized at the teacher	natural	57.73	26.46	42.09	36.29	
TRADES	–	–	–	–	–	<i>not implemented in our framework</i>
MART	–	–	–	–	–	<i>not implemented in our framework</i>
ARD	natural	57.61	20.24	38.92	29.96	
RSLAD	natural	59.68	21.30	40.49	31.40	
AdaAD	natural	59.79	23.19	41.49	33.42	
AdaAD + IGDM ($\alpha = 20$)	natural	1.00	1.00	1.00	1.00	<i>collapses</i>
AdaAD + IGDM ($\alpha = 1$)	natural	48.25	19.48	33.87	27.75	
ADR	self	–	–	–	–	<i>external repo present, not yet run</i>
CURE	self	–	–	–	–	<i>seven runs, none reproduce; see Section D</i>
RPAT++	–	55.93	27.36	41.64	36.74	
CFA (ours)[†]	natural	62.65	28.77	45.71	39.43	

Table 5: Tiny-ImageNet-200, ResNet-18.

	Method	Clean	AA	NRR	Src
	PGD-AT	45.87	18.06	25.91	ADR
(a)	Rade & Moosavi-Dezfooli	52.60	18.14	26.96	ADR
	Dong et al.	47.46	18.29	26.42	ADR
(b)	ARD / RSLAD / AdaAD / IGDM	–	–	–	<i>not reported</i>
(c)	LBGAT / ARREST / CURE	–	–	–	<i>not reported</i>
	ADR (AT + ADR)	48.55	20.23	28.61	ADR
(d)	RPAT	–	–	–	<i>not reported</i>
	CFA (ours), 200-epoch teacher[†]	55.16	20.54	29.93	ours
	CFA (ours), 80-epoch teacher [†]	57.08	18.96	28.46	ours

their own table, with each objective run twice: once as published, and once with the teacher it would actually have in our setting.

Backbone caveats. ARREST’s CIFAR-10 headline is WideResNet-34-10 (Table 6); the ResNet-18 row in Table 3 is its smaller-backbone entry. CURE reports no AutoAttack on CIFAR-100. RPAT’s WideResNet results on CIFAR-100 are PreActResNet-18. Each cell states the backbone its source used.

Reproductions. We reproduce RPAT++ to within 0.31 AutoAttack of its published numbers on both datasets and report our own measurement. CURE we could *not* reproduce: seven runs of the official repository fall short on *different* axes—the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Computing its RGP prominence mask on absolute values moves AA by +15.9, which localizes the discrepancy without closing it, so the CURE rows above are the paper’s own numbers.

4.3 PUBLISHED DISTILLATION OBJECTIVES, GIVEN A NATURAL TEACHER

Two readings, and the first is why these numbers are here rather than in Tables 3 and 4. **Changing the teacher costs these objectives 4.5 to 9 points of AutoAttack while barely moving standard**

Table 6: CIFAR-10, WideResNet-34-10.

	Method	Clean	AA	NRR	Src
(a)	PGD-AT / TRADES	–	–	–	<i>to fill</i>
(b)	ARD / RSLAD / AdaAD / IGDM	–	–	–	<i>to fill</i>
	LBGAT	88.22	52.18	65.57	ARREST
(c)	ARREST	90.24	50.20	64.51	ARREST
	CURE	87.05	52.10	65.19	RPAT
	ADR	84.67	53.25	65.38	RPAT
(d)	ReBAT	85.25	54.78	66.70	RPAT
	RPAT++	86.76	54.97	67.30	RPAT
	CFA (ours)	<i>running</i>			ours

Table 7: CIFAR-100, WideResNet-34-10.

	Method	Clean	AA	NRR	Src
(a)	PGD-AT (SAT)	53.64	21.01	30.20	F ² AT
(b)	ARD / RSLAD / AdaAD / IGDM	–	–	–	<i>to fill</i>
	LBGAT	70.25	26.73	38.72	ADR
(c)	ARREST	73.05	24.32	36.50	ADR
	ADR (AT + ADR)	62.21	31.60	41.90	ADR
(d)	F ² AT	60.21	26.91	37.23	F ² AT
	RPAT	–	–	–	<i>PreActRN-18 only</i>
	CFA (ours)	<i>running</i>			ours

accuracy—ARD 24.77 $\rightarrow$ 20.24 and 49.49 $\rightarrow$ 43.75, RSLAD 26.76 $\rightarrow$ 21.30 and 52.13 $\rightarrow$ 42.91, AdaAD 28.06 $\rightarrow$ 23.19 and 52.79 $\rightarrow$ 46.22. What those methods deliver is therefore mostly attributable to the teacher, which is why their published numbers cannot be set beside a row produced without one.

Second, **with the natural teacher every one of them lands below simply not distilling**: on both datasets, plain adversarial training initialized from the same teacher beats all four. The accuracy a natural teacher holds is not recovered by treating it as a logit target.

AdaAD + IGDM collapses to chance on both datasets at the published $\alpha = 20$ (1.00 on CIFAR-100, 10.00 on CIFAR-10), so ten classes did not rescue it. This is not a porting artifact: with a natural teacher the IGDM target $\text{softmax}(f_T(x+\delta) - f_T(x-\delta))$ places 0.917 of its mass on a single class and the term reaches $17.3\times$ the AdaAD loss. Reducing the weight to $\alpha = 1$, which puts the two terms on comparable scales and is the most favourable reading available, trains normally and still finishes far below AdaAD alone. Table 9 gives the structural reason.

Our adversarially trained network reproduces IGDM’s reported value, so the measurement is calibrated; the natural teacher is forty times further from linear, and the finite difference IGDM matches is not a gradient there.

4.4 COMPONENT ABLATION

The sensitivity rule contributes exactly +0.49 NRR at both schedule lengths (+1.61/+0.21 at 50 epochs, +1.77/+0.19 at 100) at identical total attack budget. AWP changes sign with schedule length: −0.25 NRR at 50 epochs, +1.21 at 100. And 50 epochs with weight averaging alone already exceeds ADR’s full stack (38.46 against 38.08), at half the schedule and one fewer component, so the gain is not the stack.

Leaving the teacher’s classifier alone beats every alternative, including the best temperature. At the full recipe the same holds—freezing the head scores 62.65/28.77/NRR 39.43 against 62.35/28.68/39.29 with the head-KD term retained—and this is what removes τ and β from the method.

Table 8: The four published distillation objectives, given the teacher they would have in our setting. The right block of each dataset is **measured in our framework**: ResNet-18, 100 epochs, each method at its own published recipe, same data pipeline and same evaluation as every other row we report. The left block is *quoted* from Lee et al. (2025), which runs all four with an adversarially trained WideResNet teacher under one protocol; it is printed only so the cost of changing the teacher is visible, and it is not a comparison against our own numbers. Blank cells are runs that have not finished.

	CIFAR-100				CIFAR-10			
	<i>quoted</i> : robust teacher		ours : natural teacher		<i>quoted</i> : robust teacher		ours : natural teacher	
	Clean	AA	Clean	AA	Clean	AA	Clean	AA
ARD	61.51	24.77	57.61	20.24	85.04	49.49	84.58	43.72
RSLAD	60.22	26.76	59.68	21.30	83.59	52.13	85.15	42.81
AdaAD	64.43	28.06	59.79	23.19	84.74	52.79	85.39	46.12
AdaAD + IGDM	64.44	30.32	1.00	1.00	84.83	54.02	10.00	10.00
at $\alpha = 1$	–	–	48.25	19.48	–	–	–	–
<i>the same natural teacher used for initialization only, or as an anchor</i>								
PGD-AT from the teacher	–	–	57.73	26.46	–	–	82.23	50.00
CFA (ours), no WA/AWP	–	–	61.21	25.24	–	–	–	–

Table 9: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an $\epsilon = 8/255$ ball, computed as in IGDM’s Sec. 3.1.

Network	Remainder proportion
IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published)	0.012 / 0.012 / 0.016
PGD-AT ResNet-18 (ours)	0.0111
Natural teacher (ours)	0.4763
Anchored student, $p = 0$	0.0029
Anchored student, shipped recipe	0.0202

4.5 TRAINING RADIUS

Two things hold on both datasets. NRR peaks at $8.8/255$, but the curve is mild and the standard $8/255$ already exceeds the best published result on either dataset (38.72 against ADR’s 38.08; 64.48 against CURE’s 63.19), so the non-standard radius is a refinement rather than the source of the result. And past $8.8/255$ the radius buys nothing: AutoAttack and CW are identical to the last decimal at $10/255$ while standard accuracy falls 2.66 and 2.29, matching the ϵ -independence Section A predicts above the threshold.

4.6 WHAT THE TEACHER TRANSFERS

Here $r(\text{teacher clean, student clean}) = -0.999$: the teacher gains 2.51 points across the ladder and the student *loses* 2.04. NRR peaks in the middle, so the teacher’s training length is a control on the operating point, and locating it costs five natural-training runs and no adversarial ones. The effect reproduces on Tiny-ImageNet, where it is accuracy-controlled: two teachers 0.32 apart move the student by 1.92 clean and 1.58 AA.

4.7 LARGER ARCHITECTURE

WideResNet-34-10 at the same recipe on both CIFAR datasets. (*Running; the comparison line is ADR’s WRN-34-10 row, 62.21 clean / 31.63 AA on CIFAR-100.*)

Table 10: CIFAR-100, ResNet-18, raw- ℓ_2 anchor with $\epsilon_{\text{train}} = 8.8/255$ held fixed and no learning-rate freeze. Each row adds one component to the row above.

	Clean	PGD-20	CW	AA	NRR
<i>50 epochs</i>					
Anchor ($p = 0$)	61.33	32.94	27.80	26.19	36.71
+ sensitivity-matched ϵ	62.94	32.27	28.23	26.40	37.20
+ weight averaging	61.11	34.30	29.66	28.06	38.46
+ AWP [†]	59.90	35.70	29.74	28.05	38.21
<i>100 epochs</i>					
Anchor ($p = 0$)	61.21	31.26	26.82	25.24	35.74
+ sensitivity-matched ϵ	62.98	31.30	27.05	25.43	36.23
+ weight averaging	62.43	33.49	28.96	27.39	38.08
+ AWP [†]	62.35	36.26	30.65	28.68	39.29

Table 11: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP, $\epsilon = 8/255$), so that no stack component can absorb the difference.

	Clean	PGD-20	CW	AA	NRR
Logit target only, $\tau = 1$	58.26	22.62	22.79	20.84	30.70
Logit target only, $\tau = 4$	59.39	30.30	26.84	24.48	34.67
Logit target only, $\tau = 16$	57.78	31.57	26.11	24.00	33.91
Feature anchor + head KD, $\tau = 1$	62.34	26.34	26.11	23.93	34.58
Feature anchor + head KD, $\tau = 16$	62.61	32.47	27.32	25.61	36.35
Feature anchor, head untouched	62.72	28.69	27.80	25.88	36.64
head refit, adversarial CE	62.77	29.93	27.66	25.65	36.42
head refit, adv. CE + smoothing 0.1	62.57	30.43	27.55	25.66	36.39
head refit, clean CE	62.70	28.90	27.23	25.15	35.90

A TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Section 2.3 explains why anchoring a non-robust teacher is *free*. Neither result below closes the complementary question—how much robustness the objective buys—and both are reported with the reason they fall short, because each is instructive about where a proof would have to come from.

A.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

Following Tsipras et al. (2018): a robust feature $x_1 = y$ with probability p , non-robust features $z_j \sim \mathcal{N}(\eta y, 1)$, an ℓ_∞ adversary on z only, and a natural teacher $\Phi_t = m(z)$. Take $\Phi_s = ax_1 + bm(z)$.

Proposition 2. *With the inner maximum exact, $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$ with $R = ax_1 + (b - 1)m$, kinked at $b = 0$ with subgradient jump $2\epsilon \mathbb{E}|R(a, 0)| > 0$; above a threshold ϵ_0 the minimizer has $b^* = 0$ exactly.*

The adversarial term is an ℓ_1 penalty on the non-robust coefficient: the teacher supplies the value, and the inner maximization discards the route it took there. Verified numerically at $d = 512$: $b^* = 0.510$ at 0.5η and exactly 0 at $\eta, 2\eta$ and 4η . Since every ϵ in J multiplies b , at $b^* = 0$ the objective is $\mathbb{E}(ax_1 - m)^2$ and is ϵ -free—past the threshold a larger radius costs the anchor nothing, which matches the saturation measured in Table 12, where AutoAttack and CW at $10/255$ are identical to the last decimal to those at $8.8/255$ on both datasets.

This model rules out the paper’s effect by construction, so Proposition 2 cannot be the explanation of it: its reliability spectrum has two atoms and no mass near $\eta_j/\epsilon = 1$, where the effect lives. It is a statement about the free half only.

Table 12: ϵ_{train} swept with everything else fixed. Evaluation is at $\epsilon = 8/255$ in every cell.

	Clean	PGD-20	CW	AA	NRR
<i>CIFAR-100</i>					
8/255	63.89	31.41	29.83	27.78	38.72
8.8/255 [†]	62.65	32.63	30.66	28.77	39.43
10/255 [†]	59.99	32.82	30.53	28.77	38.89
<i>CIFAR-10</i>					
8/255 [†]	87.22	52.43	53.55	51.15	64.48
8.8/255 [†]	85.58	52.94	53.88	51.79	64.53
10/255 [†]	83.29	53.25	53.88	51.79	63.87

Table 13: Only the teacher checkpoint changes; the student configuration is untouched. CIFAR-100, ResNet-18. S_w/S_b is the ratio of within-class to between-class scatter on the unit sphere.

Teacher	Teacher clean	Teacher S_w/S_b	Student clean	Student AA	NRR
50 epochs	75.81	1.100	64.28	24.38	35.35
100 epochs	76.62	0.982	63.65	25.20	36.11
150 epochs	77.52	0.887	62.93	25.78	36.51
200 epochs	77.65	0.808	62.72	25.88	36.64
300 epochs	78.32	0.712	62.24	25.80	36.48

A.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing $\gamma_t(x)$ for the teacher’s margin at x and D_y for the relevant decision-boundary constant, Cauchy–Schwarz gives that $\gamma_t(x) > D_y \cdot L(x)$ implies the student is robustly correct at x . The certificate is vacuous at the scales we train at, but the ratio DL/γ_t tracks both axes across the teacher ladder of Table 13. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

B PER-ATTACK METRICS

Table 14 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Table 14: Per-attack accuracy (%) for our cells. ResNet-18, $\epsilon = 8/255$, final-epoch weight-averaged model, full test set.

Dataset	Cell	Clean	FGSM	PGD-20	CW	AA	NRR
CIFAR-100	CFA, head frozen (shipped)	62.65	37.08	32.63	30.66	28.77	39.43
	CFA, head-KD term retained	62.35	38.80	36.26	30.65	28.68	39.29
	CFA, $\epsilon_{\text{tr}} = 8/255$	63.89	36.67	31.41	29.83	27.78	38.72
	CFA, $\epsilon_{\text{tr}} = 10/255$	59.99	36.50	32.82	30.53	28.77	38.89
	CFA, no WA / AWP	61.21	35.54	31.26	26.82	25.24	35.74
	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
CIFAR-10	CFA, $\epsilon_{\text{tr}} = 8.8/255$ (shipped)	85.58	59.77	52.94	53.88	51.79	64.53
	CFA, $\epsilon_{\text{tr}} = 8/255$	87.22	59.92	52.43	53.55	51.15	64.48
	CFA, $\epsilon_{\text{tr}} = 10/255$	83.29	58.98	53.25	53.88	51.79	63.87
	RPAT++ (our reproduction)	82.41	59.73	55.72	52.88	50.75	62.82

Why PGD-20 is not the metric here. The student inherits the teacher’s classifier and never re-trains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for a different feature distribution; AutoAttack, which includes a targeted APGD, and CW_∞ , which

attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 14 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

	PGD-20	CW	AA
head-KD term retained	36.26	30.65	28.68
head frozen	32.63	30.66	28.77
difference	-3.63	+0.01	+0.09

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20 moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

C PUBLISHED RESULTS, FOR CONTEXT

Tables 3 and 4 contain only numbers we measured ourselves. Table 15 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Table 15: Published results, ResNet-18, ℓ_∞ , $\epsilon = 8/255$. Quoted as reported.

Method	Teacher	Clean	AA	Avg	NRR	Source
<i>CIFAR-100</i>						
PGD-AT	—	56.56	25.02	40.79	34.69	RPAT
TRADES	—	55.39	24.51	39.95	33.98	RPAT
MART	—	49.83	25.00	37.41	33.30	RPAT
Consistency-AT	—	58.53	25.39	41.96	35.42	RPAT
Consistency-AT + RPAT	—	60.33	26.31	43.32	36.64	RPAT
F ² AT	—	54.19	23.24	38.71	32.53	F ² AT
Generalist++	co-tr.	62.97	23.96	43.47	34.71	Generalist++
ADR (AT + ADR)	self	56.10	26.87	41.48	36.34	ADR
ADR (AT + WA + AWP + ADR)	self	57.36	28.50	42.93	38.08	ADR
<i>CIFAR-10</i>						
PGD-AT	—	82.78	44.63	63.71	57.99	CURE
TRADES	—	82.41	48.37	65.39	60.96	CURE
MART	—	80.70	47.49	64.10	59.79	CURE
ST-AT	—	83.10	50.50	66.80	62.82	CURE
IAD	robust	80.63	50.17	65.40	61.85	CURE
Consistency-AT + RPAT	—	84.12	48.98	66.55	61.91	RPAT
Generalist++	co-tr.	89.09	46.07	67.58	60.73	Generalist++
ARREST	natural	86.63	46.14	66.39	60.21	ARREST
CURE	self	86.76	49.69	68.23	63.19	CURE
ADR (AT + ADR)	self	82.41	50.38	66.40	62.53	ADR

Read against Table 4, our CFA row (62.65/28.77, Avg 45.71) exceeds every published Avg and NRR on CIFAR-100, and on CIFAR-10 (85.58/51.79, Avg 68.69) it exceeds every published Avg and NRR there as well. We state this as a comparison across codebases and not as a controlled result, which is what Tables 3 and 4 are for.

D THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74 $\rightarrow$ 45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 15 is therefore the paper’s own number, and CURE is left empty in Tables 3 and 4.

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